

MakeLife: Entity-Relationship (ER) Diagram (Chen's Notation)

The following Entity-Relationship Diagram uses exactly the classical Chen's Notation for database modeling.

- **Rectangles** represent Entities.
- **Diamonds** represent Relationships between Entities.
- **Ovals** represent Attributes of the Entities.

```
graph TD
    %% Define Styles
    classDef entity fill:#c2f0c2,stroke:#333,stroke-width:2px,color:#000;
    classDef relationship fill:#d9d982,stroke:#333,stroke-
width:2px,color:#000,shape:diamond;
    classDef attribute fill:#e6ffe6,stroke:#333,stroke-width:1px,color:#000,shape:stadium;

    %% -----
    %% ENTITIES (Rectangles)
    %% -----
    E_User[USER]:::entity
    E_Cash[CASH DONATION]:::entity
    E_Goods[GOODS DONATION]:::entity
    E_Child[CHILD]:::entity
    E_Volunteer[VOLUNTEER]:::entity

    %% -----
    %% RELATIONSHIPS (Diamonds)
    %% -----
    R_MakesCash{makes}:::relationship
    R_MakesGoods{makes}:::relationship
    R_Applies{applies}:::relationship
    R_Adopts{requests adoption}:::relationship
    R_Dedicated{dedicated to}:::relationship

    %% Connect Entities to Relationships
    E_User ---|1| R_MakesCash ---|M| E_Cash
    E_User ---|1| R_MakesGoods ---|M| E_Goods
    E_User ---|1| R_Applies ---|1| E_Volunteer
    E_User ---|1| R_Adopts ---|M| E_Child

    E_Cash ---|M| R_Dedicated ---|1| E_Child

    %% -----
    %% ATTRIBUTES (Ovals)
    %% -----

    %% User Attributes
    A_U_Name([name]):::attribute
    A_U_Email([email]):::attribute
    A_U_Phone([phone]):::attribute
```

```
E_User --- A_U_Name
E_User --- A_U_Email
E_User --- A_U_Phone

%% Cash Donation Attributes
A_C_Id([_id]>::attribute
A_C_Amount([amount]>::attribute
A_C_Date([date]>::attribute

E_Cash --- A_C_Id
E_Cash --- A_C_Amount
E_Cash --- A_C_Date

%% Goods Donation Attributes
A_G_Id([_id]>::attribute
A_G_Items([items]>::attribute
A_G_Status([status]>::attribute
A_G_Address([address]>::attribute

E_Goods --- A_G_Id
E_Goods --- A_G_Items
E_Goods --- A_G_Status
E_Goods --- A_G_Address

%% Child Attributes
A_Ch_Id([_id]>::attribute
A_Ch_Name([name]>::attribute
A_Ch_Age([age]>::attribute
A_Ch_Story([story]>::attribute

E_Child --- A_Ch_Id
E_Child --- A_Ch_Name
E_Child --- A_Ch_Age
E_Child --- A_Ch_Story

%% Volunteer Attributes
A_V_Id([_id]>::attribute
A_V_Avail([availability]>::attribute
A_V_Status([status]>::attribute

E_Volunteer --- A_V_Id
E_Volunteer --- A_V_Avail
E_Volunteer --- A_V_Status
```